



Synergistic maturity group and planting date adaptation sustains soybean yields under warming

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Plain Language Summary

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Warmer summers can decrease U.S. soybean seed yields if agronomic management is not adapted accordingly. Our research question is: Can farmers sustain soybean yields under high temperatures by changing cultivar maturity group and sowing dates? We quantified the impacts of high temperatures on soybean yields across thousands of fields in the US Mid-South and investigated whether adapting sowing dates and maturity groups could offset these negative impacts. Soybean yields dropped by nearly 6% on average with a +2°C temperature increase. However, planting earlier and choosing a later-maturing soybean variety can increase seed yields by 12% on average under high temperatures. Maturity group selection and planting date are among the most actionable management decisions for farmers, as they can rapidly adopt these changes without incurring significant production costs. These choices can be further optimized under warming to offset yield penalties.

Response to the reviewer and editor comments

We want to thank the reviewers and editors for the thoughtful suggestions, which improved the quality of our manuscript. All the comments have been carefully analyzed and answered one by one below (our responses are in blue).

Associate Editor's Comments to Author:

The reviewers found the 1) topic of the manuscript interesting and relevant, 2) paper well written, and 3) methods well described and model calibration based on sound data. However, the reviewers note that results and conclusions have been reported, and early planting dates similar to those examined (e.g., early May), and the longer maturity groups proposed are already commonly used. So the novelty and impact of the paper are unclear and limited. Items to help strengthen study include: evaluating more scenarios, providing a mechanistic explanation for the observed yield responses, clarify novelty of study compared to similar studies/simulations (e.g., Elli et al., 2022).

Response: We appreciate the Editor's feedback. We have clarified the main advancements of this work relative to previous studies and its novelty. We have added a new figure and discussion to support a mechanistic explanation for the observed yield changes. Moreover, we have moved relevant information from the supplementary material to the main text and added a new figure to quantify phenological changes driven by high temperatures and adaptive scenarios. Instead of exploring additional scenarios, we have modified the existing ones to reflect a more extreme yet realistic condition.

Technical Editor's Comments to Author:

As the AE noted, the reviewers did not find fault with the rigor or soundness of this work. Rather, the concerns were primarily related to novelty and impact of the manuscript. With this in mind, I suggest the authors consider reframing the context of the manuscript to highlight better the originality of the work and the knowledge gap that it is addressing. The topic is certainly of interest to AEL's readership, so if the reworked manuscript still falls within the journal's word limits, I would invite you to resubmit for another round of peer review.

Response: We appreciate the technical editor's feedback. As noted in the previous comment, we have reworked the manuscript to address previous reviewers' suggestions and highlight its novelty and impact. The main changes included: 1) reframing the context and title to reflect the originality of the work and the knowledge gap, 2) providing a mechanistic explanation for the observed yield changes, and 3) explaining the main advancements compared to previous studies. To the best of our knowledge, this is the first paper to quantify the interactive and synergistic effects of sowing date and genotype maturity selection as a strategy to adapt to high temperatures. An additional advance is the scale of our simulations, which covered over 4,000 fields and 40 weather years. Lastly, we have tested and applied the newest APSIM NG soybean model, which currently lacks testing studies.

Reviewer 1:

Comment #1

The authors have conducted a simulation study to evaluate the impacts of earlier planting and variety selection for soybean in the Mid-South to offset heat and drought stress. The paper is short with minimal content. Similar results and conclusions have been reported in the literature, so the novelty and impact of the paper are minimal. Authors could perhaps expand the message to increase the paper's novelty. There is plenty of room for expansion to a fuller, more comprehensive research paper.

Response: Thank you for the feedback. We believe our contribution is original for the following reasons:

- This is the first study to quantify the synergistic impact of maturity group and sowing date under possible warming scenarios. The title was reframed to reflect its novelty better. This was highlighted in the introduction:

“While farmers have been shifting towards earlier planting and maturity groups well-adapted to a region, the synergistic effect of these shifts under a warming scenario remains unquantified.”

- Our study is unique in terms of resolution of simulations, quantifying impacts across over 4,000 fields and 40 weather years per field. We have also moved relevant graphs from the supplementary material to the main document (Figure 1) to reinforce all simulations performed. The modifications are highlighted below:

Introduction

“We conducted a comprehensive simulation study across 4,651 fields and 40 weather-years using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. We considered soybean production systems in the Arkansas Delta region as a case study, an environment particularly vulnerable to warming (Sharma et al., 2023), with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025).”

Results

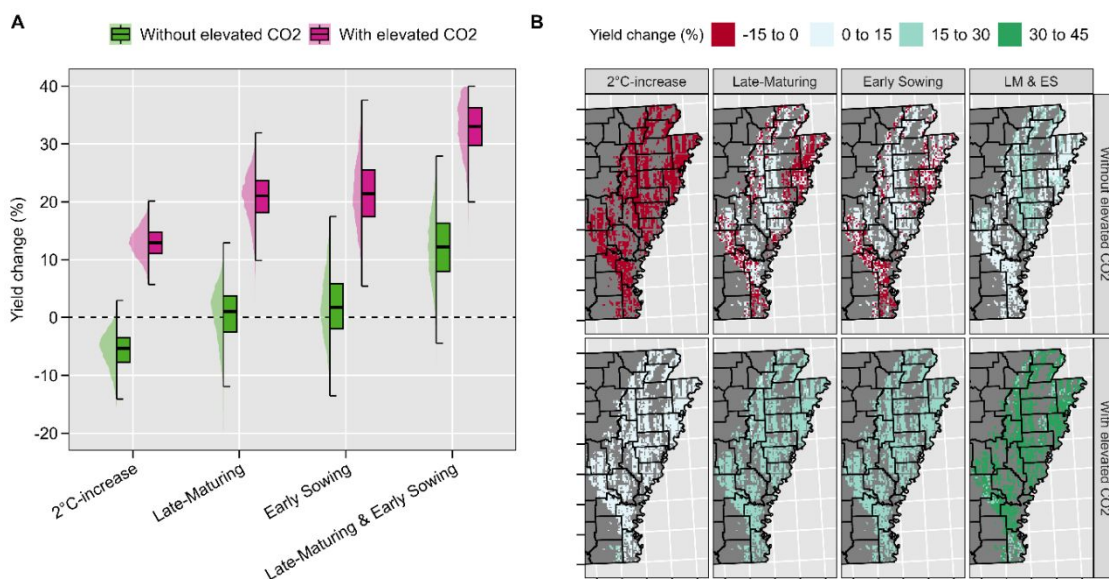


Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

- We advanced previous simulation studies (e.g., Elli et al. 2022) by exploring stacked management options under warming. We have added the following sentences to the discussion to clarify the main advances compared to previous studies:

“Our study extends previous work (Elli et al., 2022; Nendel et al., 2023) by quantifying the synergistic effect of stacked agronomic management strategies to mitigate heat risks in irrigated soybeans at a fine spatial resolution in the U.S. Mid-South.”

“We found a 5.7% decrease in soybean yield due to rising temperatures, with values ranging from -14.7% to -2.5% across sites (Figure 1). Elli et al. (2022) reported an average 19% reduction in seed yield at 10 locations in the US Corn Belt under a +2°C temperature scenario without adaptive strategies. While Elli et al. (2022) did not quantify the combined impact of high temperatures and stacked agronomic management strategies, they found that sowing 12 days earlier increased soybean seed yield by 4.7%.”

- We have tested and applied the newest APSIM framework (APSIM Next Generation), which advances previous APSIM classic work. This is highlighted in Material and Methods:

“The APSIM-Soybean model has previously proven successful in simulating the growth of various soybean maturity groups under different environmental conditions in the US (Archontoulis et al., 2020; Elli et al., 2022). We further tested the model against

experimental data and updated cultivar coefficients to the latest APSIM Next Generation (Holzworth et al., 2018) version 2025.3.7681.”

Comment #2

L54-64 – Minimize use of first person pronouns here and throughout.

Response: Partially addressed. We appreciate the reviewer's suggestion, but we believe that in some cases, the active voice improves readability, which is accepted by the scientific community.

Comment #3

L57 – There appears a conflicting logic with L47-49. It is stated that earlier planting shifts the crop cycle to cooler periods, reducing exposure to heat and drought. But here, the hypothesis proposes later maturity group soybeans, which would extend the growing season. Please clarify the reasoning for the hypothesis and for the use of later maturity group soybeans to avoid heat and drought.

Response: Thank you for the important comment. Our hypothesis is that a late-maturing variety, combined with early planting, will offset the negative impacts of warming. We agree with the reviewer that late-maturing genotypes would extend the growing season, but early planting could help reduce the occurrence of extreme weather events during the crop's critical period. Moreover, under warming, development rates will accelerate, reducing biomass production, and late-maturing genotypes can help offset the negative impacts of accelerated development rates. Warming will also increase the frequency of heat and drought events. Therefore, the net effects on soybean yield will depend on the site-specific tradeoffs and synergies between cycle-length alterations and increased occurrence of environmental stress.

Comment #4

L84 – What does it mean that the Cropland Datalayer was used to define the fields? Define what about the fields?

Response: Thank you for the suggestion, and we have clarified this. The new sentence reads: *“We used the soybean cropland data layer (NASS, 2025) to identify soybean-cropped pixels and define the spatial extent of the simulation fields, which were resampled to a 2.5 km grid (~0.025°). Each grid cell classified as soybean cropland was treated as one simulation field.”*

Comment #5

L110 – What does “advancing sowing date by 12 days” mean? Earlier or later?

Response: Thank you for the suggestion. We have clarified it. Moreover, we have altered the sowing date scenario to reflect more contrasting yet realistic sowing dates based on the USDA-NASS sowing date progress. We selected the 50% progress as the baseline and the 10% progress as the early sowing. The new sentences read:

“The adaptive scenarios included (1) an early MG 5 cultivar instead of early MG 4, (2) an early sowing date by 4 weeks (April 24), corresponding to the average USDA-NASS 10% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4), and (3) a combination of an early MG 5 cultivar with early sowing.”

We have added a new supplementary figure to facilitate understanding of the choices in the scenario analysis:

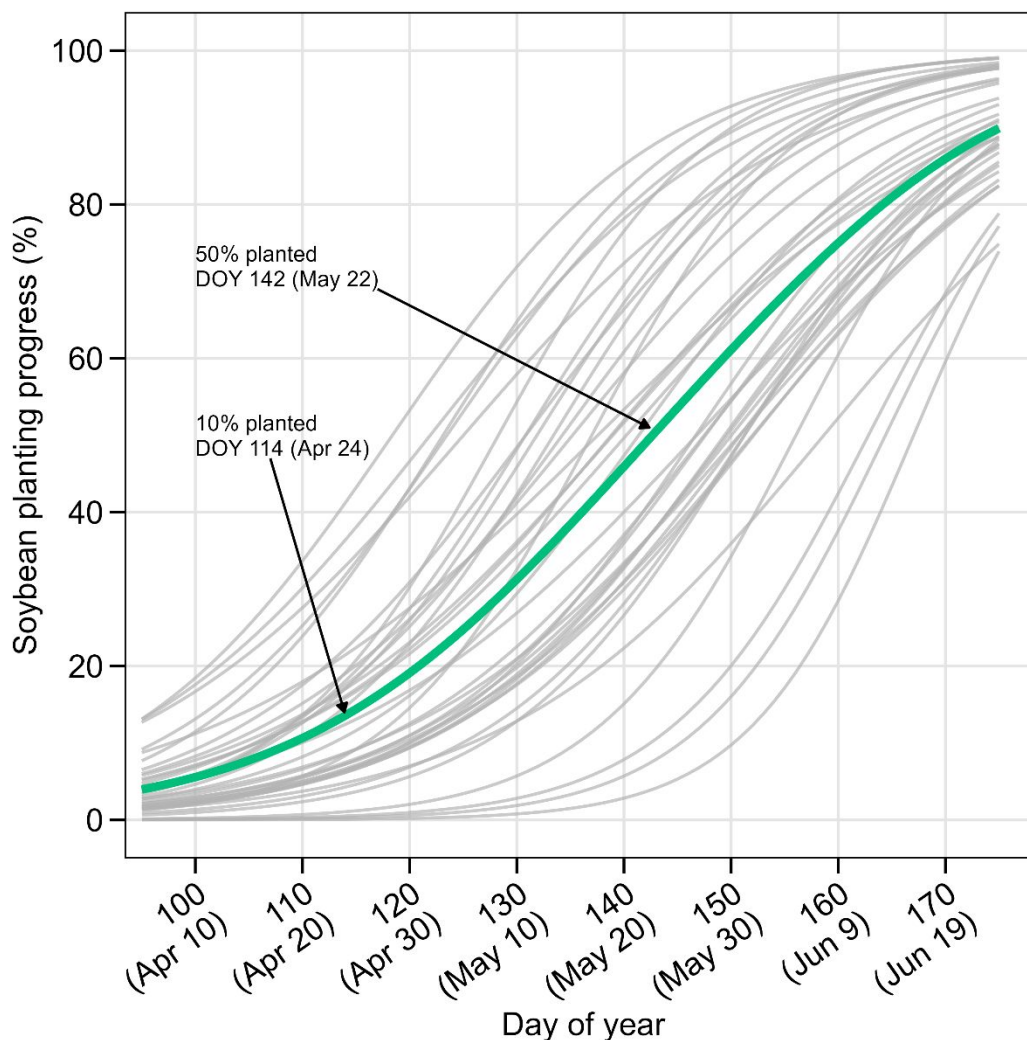


Figure S4. USDA-NASS soybean sowing progress in Arkansas from 1990 to 2025. Grey lines represent the historical years, while the green, thicker line indicates the average sowing progress. 50% and 10% average sowing progress are indicated in the figure.

Comment #6

L186 – Discuss the limitations of the study in terms of uncertainty in the model structure or model input data.

Response: Addressed. We have expanded the discussion of structural model uncertainties. The new text reads:

“While elevated [CO₂] can offset the negative impacts of rising temperatures (Ainsworth et al., 2002b), genotypic differences in [CO₂] response may exist (Bishop et al., 2015). Moreover, structural model uncertainties exist in predicting yield benefits under elevated [CO₂] (Kothari et

al., 2022). Recognizing that soybeans are already a highly heat-tolerant crop, heat events above 28°C during the reproductive phase can reduce yield by lowering seed number and weight (Boote et al., 2005; Djanaguiraman et al., 2013). The APSIM-soybean version used in the present study accounts for temperature effects on crop growth rate (Kothari et al., 2024), but does not account for the impacts of extreme heat on seed number determination, a common limitation in process-based models (Richetti et al., 2025). Moreover, APSIM assumes constant temperature responses throughout the life cycle, whereas in soybeans this response.”

Comment #7

L194 – Overall, the paper is concise, and the content is minimal. The study provides new information on soybeans in the Mid-South, but the conclusions are not particularly novel or earth-shattering. How could the paper be expanded to be more impactful?

Response: Thank you for your feedback. We altered the text to better reflect the originality of our study. We have also explained the main advancements compared to previous studies. To the best of our knowledge, synergistic effects of sowing date and genotype maturity selection have not been quantified in previous literature. We believe our study conveys an important message for advancing soybean resilience under warming. While the reviewer states that the content is minimal, we have moved relevant information from the supplementary material, added new information on phenology and a mechanistic explanation of yield changes, and focused on the most important paper messages within the recommended word limit (2,500 words), as per the journal guidelines. We refer to comment #1 for the modified text.

Reviewer 2:

Comment #8

The paper is well written and the objectives are clearly stated. The methods are well described and model calibration is based on sound data. Validation use a wide range of environments. The results are well presented.

Response: Thank you for the positive feedback.

Comment #9

However, it would be interesting to learn a little more on the background of the yield effects, e.g., how temperature increase, early sowing and late maturity class affect the length of the seed filling period. Since soybean is responsive to short days, it would be good to know if APSIM considers any photoperiod effect and if this differs in the parameters of the different maturity groups.

Response: Thank you for the thoughtful suggestion. We have created a new figure illustrating phenological changes under different scenarios, including the entire crop cycle and the seed-filling period. APSIM accounts for photoperiod effects on development, as evidenced by the calibrated photoperiod modifier parameters across maturity groups in Table S1.

The new figure is shown below:

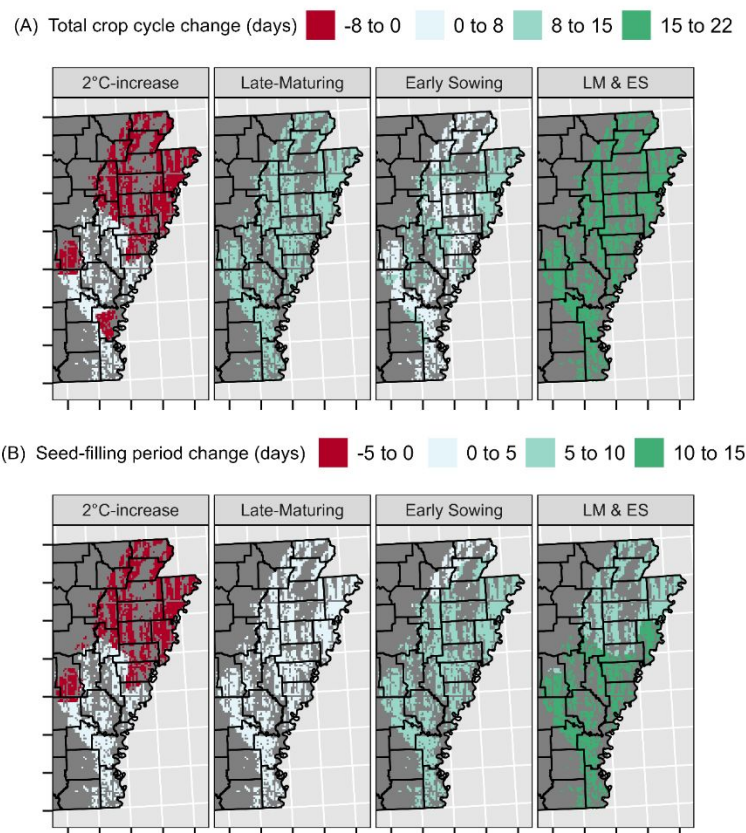


Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

The new text expanding on the background of the yield effect reads:

Results

“A shortening of the total crop cycle was observed in 61% of the sites, with more pronounced changes in northern environments (Figure 2), where baseline temperatures are lower. The total crop cycle length decreased by up to 15 days across all site-years. The seed filling period was shortened by up to 9 days across all site-years.”

“The combined adaptive scenario extended the seed-filling period by an average of 10.4 days (Figure 2).”

Discussion

“Extending the seed filling period was a promising strategy to maximize soybean seed yield (Figure 1), which agrees with the findings of Boote et al. (2003). However, phenology did not fully explain yield reduction under warming, especially in the Southern region (Figures 1 and 2). This could be associated with slower development on days when temperatures exceed the optimum temperature for soybean development in APSIM (30°C, Figure S3). Battisti et al. (2018) observed that, owing to the temperature response function governing phenological development, the APSIM model simulated a greater reduction in soybean cycle length under a +4.5°C warming scenario than under a +6.0°C scenario in Southern Brazil, with a mean air temperature 23.65°C. Further yield decreases were likely associated with high-temperature effects on radiation use efficiency (Kothari et al., 2022; Ruiz-Vera et al., 2013). While exploring optimum planting date windows for target environments was not the focus of this study, future research could investigate more extreme scenarios to enhance mechanistic understanding of yield responses under warming.”

Comment #10

line 156: please add "+" in front of "2°C scenario"

Response: Addressed and thank you.

Comment #11

Supplement figure S3: please mention if these values are without adaptation measures.

Response: Addressed. The figure refers to the baseline conditions without adaptation measures.

Comment #12

The references of NASS (2023) and of Salmerón et al. 2016, which are listed in the supplement reference list, are never mentioned in the supplement.

Response: Thank you for catching this. Addressed.

Reviewer 3:

Comment #13

The topic of the manuscript is interesting; however, the results lack novelty. The study emphasizes the need for accountable solutions to increasing temperatures in the U.S. Mid-South. To address this, a calibrated APSIM model is used to simulate soybean yield under future climate scenarios (+2°C increase in mean temperature and/or elevated CO₂), considering current and earlier planting dates in combination with changes in maturity group (MG). While the topic is relevant, the findings are not new. Farmers in the region are already adopting early planning dates similar to those explored in this study (e.g., early May). Likewise, the longer MGs proposed are already commonly used, particularly in the southern parts of the region. Additionally, the manuscript does not provide a clear explanation or mechanism underlying the simulated yield increases across scenarios. Important variables—such as the photothermal quotient during critical periods—are not explored. Evaluating more extreme scenarios and, importantly, providing a mechanistic explanation for the observed yield responses would strengthen the manuscript and improve its relevance for future readers.

Response: We appreciate the reviewer's feedback. Our scenario analysis reflects agronomic management that is realistic and actionable for the farmers in the region. That is why we use practices already adopted. The novelty of the study centers on the synergistic effects of maturity group and sowing date under warming. We have added new text to clarify it (see comment #1). Thank you for the suggestion of adding the photothermal quotient analysis. However, this goes beyond the scope of the present study. Yet, we have enhanced the mechanistic explanation of the observed yield responses by adding a new figure and discussion of changes across the whole crop cycle and the seed-filling period (see comment #9). Regarding the more extreme scenario analysis, we modified the sowing date scenario by planting earlier by 4 weeks, corresponding to the 10% sowing progress of the region, an extreme but yet realistic change (see comment #5). Results change in magnitude but not in direction and the main messages and findings remain the same.

We have added a new sentence suggesting a more detailed scenario analysis in future studies. The new sentence reads:

“While exploring optimum planting date windows for target environments was not the focus of this study, future research could investigate more extreme scenarios to enhance mechanistic understanding of yield responses under warming.”

Comment #14

Lines 50–53: This is an important point: early planting is not always optimal, as there is a planting window that maximizes radiation capture and varies with MG. However, only one “early” planting date is evaluated in this study. This does not allow identification or exploration of an optimal window, or at least this is not demonstrated.

Response: Thank you for the suggestion. As clarified in comment # 13, identifying an optimal sowing window was not the focus of this study. Our goal was to investigate whether there was a synergistic effect of planting early and using a late-maturing variety under a warming scenario.

Comment #15

Line 64: Please include a brief introduction on the role and projected changes of CO₂, as this is currently missing.

Response: Addressed and thank you. The new sentence reads:

“While projected increases in [CO₂] can partially offset heat-induced yield losses (Ainsworth et al., 2002a; Kothari et al., 2022), complementary agronomic adaptation may be needed in some environments to fully offset the yield penalty (Makowski et al., 2020).”

Comment #16

Lines 75–76: Is MG5 considered “late” in this context? Farmers are already planting MG5 and even MG6 in the southern parts of the region.

Response: Thank you for the comment. MG 5 is considered late relative to the current plausible range adopted by farmers in the region (MG 3 to MG 5). While MG6 varieties were planted in the past, they are currently not very common, and early MG4s have been the most common. The rationale for this study is that, with warming, switching back to a slightly late-maturing variety, combined with earlier planting, will generate yield advantages.

Comment #17

Lines 110–113: Consider revising the second scenario description to: “advancing the sowing date by 12 days, corresponding to an early but realistic planting window based on NASS planting progress data for the studied region (NASS, 2023)”. The current reference to the “upper tail” suggests delayed planting. Also, please clarify what is meant by “realistic” (e.g., 10% vs. 30% planting progress). Would exploring even earlier, less common planting dates provide additional insight?

Response: Thank you for the important comment. We have clarified the rationale for the chosen scenarios. Also, we moved the early-sow scenario to a more extreme situation (10% of the NASS planting progress, about 4 weeks earlier). A figure was added to the supplementary material to facilitate visualization of the selected scenarios. Detailed modifications are explained in comment #5.

Comment #18

Lines 156–158: When stating that “our study extends previous work,” please provide appropriate references.

Response: Addressed. The new sentence reads:

“Our study extends previous work (Elli et al., 2022; Nendel et al., 2023) by quantifying the synergistic effect of stacked agronomic management strategies to mitigate heat risks in irrigated soybeans at a fine spatial resolution in the U.S. Mid-South.”

Comment #19

Lines 163–164: Clearly state what is novel in this study compared to similar simulations (e.g., Elli et al., 2022).

Response: Addressed, thank you for the suggestion. Elli et al. (2022) used a previous version of APSIM (APSIM classic) with a focus on nitrogen dynamics at 10 locations in the Corn Belt (only 1 in the mid-south) and evaluated adaptive strategies only under baseline conditions, not under warming. Moreover, the study did not evaluate stacked management options. In addition to these advancements, the present study conducts fine-scale simulations (4,000+ fields) focused on the Mid-South, using the latest APSIM framework (APSIM Next-Generation). The region is predominantly irrigated with soybeans, lacks adaptive modeling studies, and is particularly vulnerable to climate change. The new text reads:

“We found a 5.7% decrease in soybean yield due to rising temperatures, with values ranging from -14.7% to -2.5% across sites (Figure 1). Elli et al. (2022) reported an average 19% reduction in seed yield at 10 locations in the US Corn Belt under a +2°C temperature scenario without adaptive strategies. While Elli et al. (2022) did not quantify the combined impact of high temperatures and stacked agronomic management, they found that sowing 12 days earlier increased soybean seed yield by 4.7%.”

Additional modifications relevant to this comment can be found in comment #1.

Comment #20

Line 173: When discussing model uncertainties, please also consider uncertainties in phenological predictions. APSIM is somewhat limited mechanistically compared to other models—for example, it assumes a constant temperature response throughout development, whereas in soybean this response changes across stages. This limitation and its potential impact on results should be discussed.

Response: Thank you for the suggestion. We have added a sentence acknowledging this limitation. The new text reads:

“Moreover, APSIM assumes constant temperature responses throughout the life cycle, whereas in soybeans this response may change across developmental stages (Hatfield et al., 2011).”

Comment #21

The optimization of phenology parameters is based on days, which may introduce bias. Using development rates would be more appropriate.

Response: Thank you for the suggestion. As shown in Figure S1, the bias is nearly zero for phenology. Thus, we believe our optimization method is consistent and does not introduce bias. Moreover, we have followed a similar approach to other studies that successfully calibrated the APSIM model, such as [Pinto et al. \(2026\)](#) and [Archontoulis et al. \(2014\)](#).

Comment #22

Figure S1: Consider plotting days after sowing instead of day of year (DOY) to better evaluate model accuracy across environments, especially for earlier planting dates. It is unclear whether the model performance under early planting was validated or simply assumed. Additionally, differentiating key stages (R1, R5, R7) with distinct colors would improve clarity.

Response: Thank you for the suggestions. We have replaced DOY with days after sowing. Regarding the suggestion to differentiate stages, we have tried this approach, and given the large number of data points, the graph becomes very hard to follow. In this context, we kept the graph layout as it was. The updated figure with days of sowing instead of DOY is shown below:

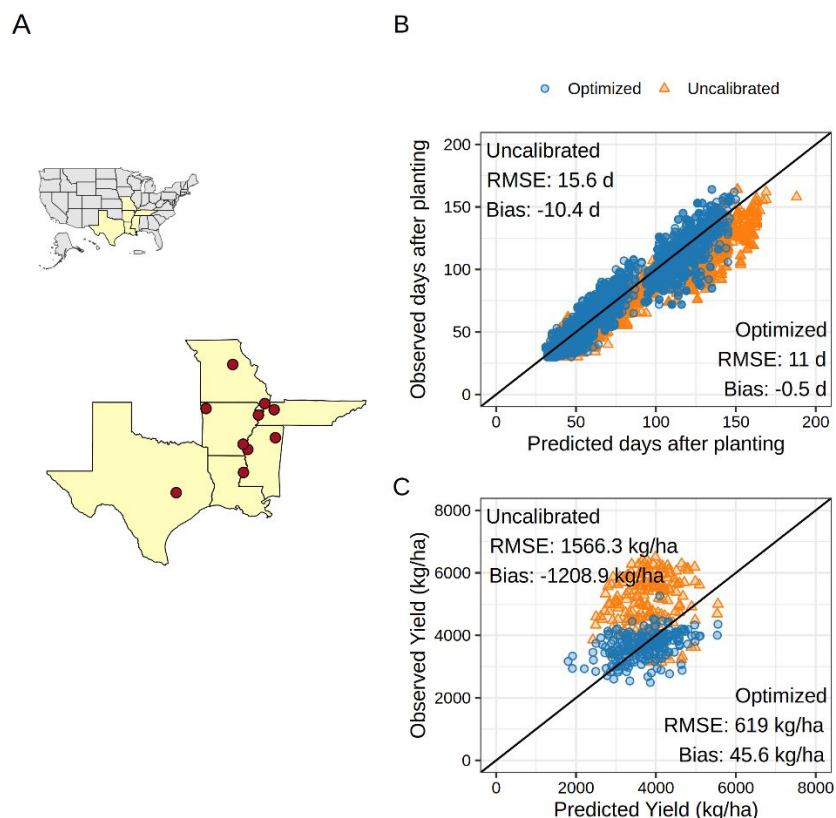


Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed days after sowing for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Comment #23

The reported RMSE (nearly two weeks) after optimization seems high. How does this compare with similar studies?

Response: Thank you for the suggestion. Given the extensive dataset and the large variability in environmental conditions (sowing dates, genotypes, locations, years; see [Salmerón et al., 2016](#)), we believe the RMSE is acceptable and within the ranges identified in the literature. We have added a new sentence to address this suggestion:

“The calibration results confirmed that the model is robust across a wide range of environmental conditions (Figures S1 and S2 and Table S1), with root mean square errors within similar ranges to those reported in other studies for soybean phenology and yield (Archontoulis et al., 2014; Salmerón and Purcell, 2016; Salmerón et al., 2017).”

Comment #24

Lines 41–42: Please provide a supporting reference.

Response: Thank you and addressed. The new sentence reads:

“We conducted a comprehensive simulation study across 4,651 fields and 40 weather-years using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. We considered soybean production systems in the Arkansas Delta region as a case study, an environment particularly vulnerable to warming (Sharma et al., 2023), with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025).”

Comment #25

Figure 1 caption: Clarify that all maturity groups (including late-maturing) also include the +2°C scenario.

Response: Thank you and addressed. We have added a new sentence to clarify that all scenarios, except the baseline, were simulated under warming. The new figure caption reads:

“Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.”

Comment #26

There appear to be two separate reference lists. Please check and correct.

Response: Thank you for the suggestion. We believe the second reference list you are referring to is probably from the supplementary material document.

**SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION
SUSTAINS SOYBEAN YIELDS UNDER WARMING**

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ABSTRACT

High temperatures increasingly threaten U.S. soybean seed yields, yet actionable solutions remain limited. We evaluated the APSIM model against observed regional data from the U.S. Mid-South and conducted gridded simulations across 4,651 fields to identify how concurrent changes in planting date and maturity group affect yield under rising temperatures and elevated [CO₂]. The +2°C temperature scenario reduced yields by 5.7% on average. However, combining a later-maturing soybean genotype with earlier sowing synergistically increased soybean yield by 12% under the +2°C scenario. These results revealed an opportunity to sustain soybean yield through management and genotype selection under warming. Advancing our understanding of management and genotypic effects on soybean phenology and improving their representation in process-based models are therefore essential for developing adaptive strategies that can support durable yield gains under high-temperature conditions.

1 INTRODUCTION

Rising temperatures and changes in rainfall patterns can decrease U.S. soybean seed yields by 19% in the absence of adaptive strategies (Elli et al., 2022). Higher temperatures accelerate phenological development, shortening the crop cycle and limiting biomass accumulation and yield (Elli et al., 2022; Rose et al., 2016). Previous simulation studies indicate that adjusting soybean phenology may help offset yield reductions under global warming (Yang et al., 2025; Zabel et al., 2021). While projected increases in [CO₂] can partially offset heat-induced yield losses (Ainsworth et al., 2002a; Kothari et al., 2022), complementary agronomic adaptation may be needed in some environments to fully offset the yield penalty (Makowski et al., 2020).

A relatively simple and well-established strategy to adjust the soybean life cycle to the available growing season length is to select soybean maturity groups based on local temperature and photoperiod conditions (Mourtzinis and Conley, 2017). Moreover, earlier planting extends the growing season and shifts the crop cycle into cooler periods, reducing exposure to late-season drought and heat events (Edwards et al., 2003; Bastidas et al., 2008). While farmers have been shifting towards earlier planting and maturity groups well-adapted to a region, the synergistic effect of these shifts under a warming scenario remains unquantified.

We conducted a comprehensive simulation study across 4,651 fields and 40 weather-years using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. We considered soybean production systems in the Arkansas Delta region as a case study, an environment particularly vulnerable to warming (Sharma et al., 2023), with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025). We hypothesize that under high temperatures, using late-maturing soybean genotypes combined with early planting will offset the negative impacts of warming on

soybean yield. Our objective was to quantify the individual and synergistic impacts of changes in maturity group and sowing date on adapting soybean production to high-temperature risks under current and elevated [CO₂].

2 MATERIAL AND METHODS

2.1. APSIM-Soybean model testing

The soybean model within APSIM Next Generation (Holzworth et al., 2018) was evaluated against regional data from the U.S. Midsouth covering multiple locations, planting dates, and cultivar maturity groups (Figure S1), as described in Salmerón et al. (2016). The APSIM-Soybean model has previously proven successful in simulating the growth of various soybean maturity groups under different environmental conditions in the US (Archontoulis et al., 2020; Elli et al., 2022). We further tested the model against experimental data and updated cultivar coefficients to the latest APSIM Next Generation (Holzworth et al., 2018) version 2025.3.7681. Cultivar coefficients were calibrated for an early maturity group (MG) 4 (relative MG 4.0 to 4.4) and an early MG 5 (rMG from 5.0 to 5.4) cultivar. The calibration results confirmed that the model is robust across a wide range of environmental conditions (Figures S1 and S2 and Table S1), with root mean square errors within similar ranges to those reported in other studies for soybean phenology and yield (Archontoulis et al., 2014; Salmerón and Purcell, 2016; Salmerón et al., 2017).

2.2. Regional simulations and scenario analysis

We conducted a simulation exercise across 40 weather years and 4,651 fields in the US Mid-South. Further information on the weather conditions of the study area is presented in Figure S3. We used the soybean cropland data layer (NASS, 2025) to identify soybean-cropped pixels and

define the spatial extent of the simulation fields, which were resampled to a 2.5 km grid (~0.025°). Each grid cell classified as soybean cropland was treated as one simulation field. We used the *apsimx* R package (Miguez, 2020) to run the simulations. Daily historical weather data on maximum and minimum temperatures, solar radiation, and rainfall were retrieved from the nearest weather station in the National Weather Service Network using the *apsimx* R package. [CO₂] was set at 350 ppm. Soil profile information was derived from SSURGO (Soil Survey Staff, 2023). Within each simulated grid cell, we used the most predominant soil profile. Soil water and inorganic nitrogen initial conditions were reset on January 1 each year and were held constant across locations to avoid confounding effects of soil vs. weather (Elli and Archontoulis, 2023). On the first day of the simulation (January 1), soil water content was set at field capacity, total inorganic nitrogen over the soil profile was 50 kg N ha⁻¹, and maize residue on the soil surface was 2000 kg ha⁻¹ with a C:N ratio of 70.

For baseline simulations, the sowing date was May 22, an average value for the studied region based on USDA-NASS 50% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4). We used an early MG 4 cultivar, which is well adapted to the region (Salmerón et al., 2016; Salmerón et al., 2014). Soybean was irrigated as it is the predominant system across the studied region (NASS, 2025). Automatic irrigation was triggered between R1 and R7 when soil water reached 50% of the plant-available water in the 0-60 cm depth, to refill soil moisture to field capacity.

For the high-temperature scenario, we considered a 2°C increase (+2°C) in daily maximum and minimum temperatures, which is an average projection derived from 15 global circulation models for 2050, as reported by Elli et al. (2022). A +2°C scenario combined with an elevated [CO₂] to 540 ppm (Elli and Archontoulis, 2023) was also evaluated. We employed a simpler

approach, rather than the direct outputs of global circulation models, because it has been shown to yield results with similar direction and magnitude while substantially reducing computational requirements (Elli et al., 2022). The adaptive scenarios included (1) an early MG 5 cultivar instead of early MG 4, (2) an early sowing date by 4 weeks (April 24), corresponding to the average USDA-NASS 10% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4), and (3) a combination of an early MG 5 cultivar with early sowing. Data analysis and visualization were conducted using R version 4.1.1 (R Development Core Team, 2024).

3 RESULTS

3.1. Elevated temperature and [CO₂] impacts on soybean yields without adaptation

The +2°C temperature scenario decreased soybean seed yields by 5.7% on average (~245 kg ha⁻¹), with values ranging from -14.7% to -2.5% across the 4,651 sites (Figure 1). The mean simulated yield for the baseline condition was 4,300 kg ha⁻¹ with a south-to-north geographical gradient ranging from 2,142 to 4,837 kg ha⁻¹ (Figure S5). Elevated [CO₂] consistently offset the negative effects of higher temperatures across sites (Figure 1), with an average yield 12.9% greater than under baseline conditions, and values ranging from 9.2% to 15.4% across all sites (Figure 1). A shortening of the total crop cycle was observed in 61% of the sites, with more pronounced changes in northern environments (Figure 2), where baseline temperatures are lower. The total crop cycle length decreased by up to 15 days across all site-years. The seed filling period was shortened by up to 9 days across all site-years.

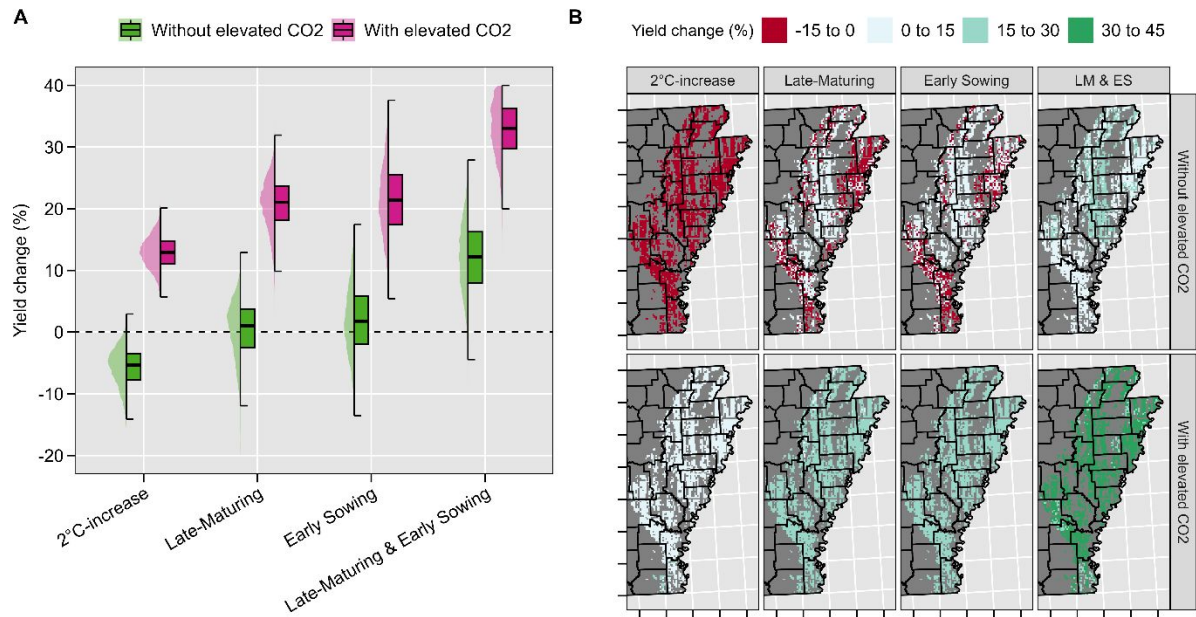
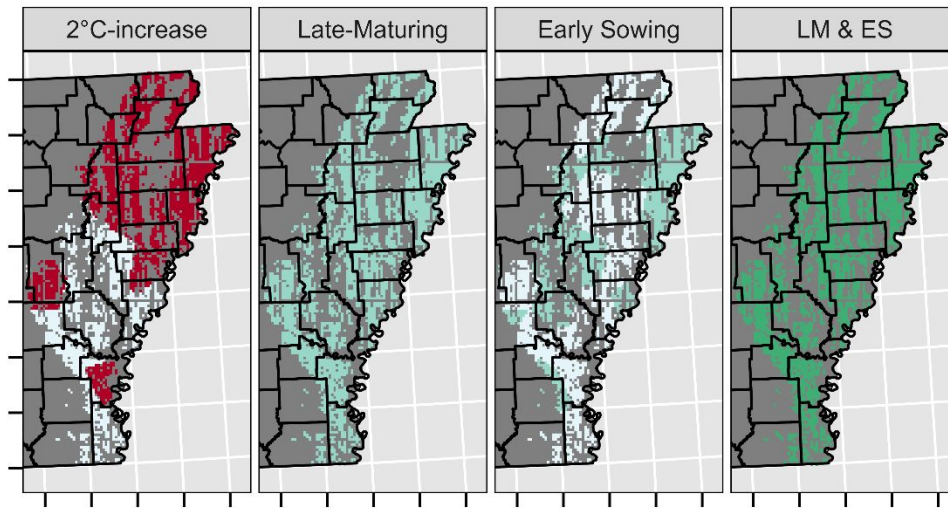


Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

(A) Total crop cycle change (days) ■ -8 to 0 ■ 0 to 8 ■ 8 to 15 ■ 15 to 22



(B) Seed-filling period change (days) ■ -5 to 0 ■ 0 to 5 ■ 5 to 10 ■ 10 to 15

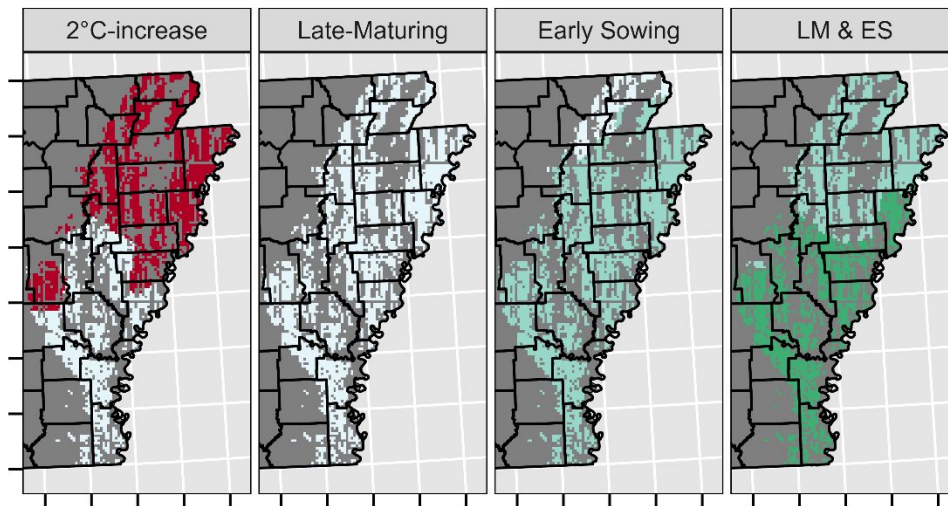


Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

3.2. Adaptive strategies to elevated temperature

Stacked agronomic management strategies were more effective than standalone strategies to maximize soybean yields under warming (Figure 1). Using a late-maturing soybean cultivar in combination with early sowing increased soybean seed yield by 12% under the +2°C temperature scenario, without elevated CO₂. The combined adaptive scenario extended the seed-filling period by an average of 10.4 days (Figure 2). Overall, the standalone strategy of using a late-maturing cultivar under a +2°C temperature scenario resulted in yields comparable to the historical baseline, with values ranging from -12.6% to 4.8% across fields. This practice increased yields at 62% of the 4,651 simulated sites, particularly in the central-north region (Figure 1B). The Early sowing strategy increased yields by an average of 2% relative to the baseline, with values ranging from -4.9% to 7.8%. This practice increased yields at 71% of the sites. Elevated [CO₂] altered the magnitude but not the direction of the yield response to stacked adaptive management options. The combined impact of elevated CO₂ and stacked agronomic management strategies increased yields by an average of 34%, with values ranging from 26.1% to 41.8% (Figure 1).

4 DISCUSSION

Our study revealed that soybean yields under warming can be sustained by concurrently adjusting cultivar maturity and planting date. Under the +2°C temperature scenario, the standalone strategies of using a late-maturing cultivar and planting earlier increased yield by an average of 0.5% and 2%, respectively, relative to the historical simulations. This implies that individual strategies offset the impacts of higher temperature, resulting in yields comparable to baseline yields under current weather conditions. However, the offset was not consistent across all sites (Figure 1). The combined strategies increased seed yield by an average of 12% in relation to the baseline, with values ranging from 0.4% to 17.6% across sites, demonstrating a

synergistic impact and emerging as the most promising approach to sustain soybean yield under a +2°C scenario. Our study extends previous work (Elli et al., 2022; Nendel et al., 2023) by quantifying the synergistic effect of stacked agronomic management strategies to mitigate heat risks in irrigated soybeans at a fine spatial resolution in the U.S. Mid-South.

We found a 5.7% decrease in soybean yield due to rising temperatures, with values ranging from -14.7% to -2.5% across sites (Figure 1). Elli et al. (2022) reported an average 19% reduction in seed yield at 10 locations in the US Corn Belt under a +2°C temperature scenario without adaptive strategies. While Elli et al. (2022) did not quantify the combined impact of high temperatures and stacked agronomic management strategies, they found that sowing 12 days earlier increased soybean seed yield by 4.7%. Deryng et al. (2014) investigated the effects of heat stress on soybeans worldwide and predicted a 33% yield reduction by 2080. Nendel et al. (2023) projected that heat stress will increasingly outweigh drought stress as a constraint to soybean production in Europe.

While elevated [CO₂] can offset the negative impacts of rising temperatures (Ainsworth et al., 2002b), genotypic differences in [CO₂] response may exist (Bishop et al., 2015). Moreover, structural model uncertainties exist in predicting yield benefits under elevated [CO₂] (Kothari et al., 2022). Recognizing that soybeans are already a highly heat-tolerant crop, heat events above 28°C during the reproductive phase can reduce yield by lowering seed number and weight (Boote et al., 2005; Djanaguiraman et al., 2013). The APSIM-soybean version used in the present study accounts for temperature effects on crop growth rate (Kothari et al., 2024), but does not account for the impacts of extreme heat on seed number determination, a common limitation in process-based models (Richetti et al., 2025). Moreover, APSIM assumes constant

176 temperature responses throughout the life cycle, whereas in soybeans this response may change
177 across developmental stages (Hatfield et al., 2011).

178 Warmer springs and longer frost-free periods have allowed farmers to plant earlier every year
179 (Mourtzinis et al., 2019). However, this practice is not always feasible due to extreme rainfall
180 events, which are projected to intensify further (Elli et al., 2022). Given the importance of
181 adapting the soybean life cycle to temperature changes, further research is needed to enhance the
182 fundamental understanding of soybean phenology and the genetic and management controls.

183 Extending the seed filling period was a promising strategy to maximize soybean seed yield
184 (Figure 1), which agrees with the findings of Boote et al. (2003). However, phenology did not
185 fully explain yield reduction under warming, especially in the Southern region (Figures 1 and 2).
186 This could be associated with slower development on days when temperatures exceed the
187 optimum temperature for soybean development in APSIM (30°C, Figure S3). Battisti et al.
188 (2018) observed that, owing to the temperature response function governing phenological
189 development, the APSIM model simulated a greater reduction in soybean cycle length under a
190 +4.5°C warming scenario than under a +6.0°C scenario in Southern Brazil, with a mean air
191 temperature 23.65°C. Further yield decreases were likely associated with high-temperature
192 effects on radiation use efficiency (Kothari et al., 2022; Ruiz-Vera et al., 2013). While exploring
193 optimum planting date windows for target environments was not the focus of this study, future
194 research could investigate more extreme scenarios to enhance mechanistic understanding of yield
195 responses under warming.

5 CONCLUSION

This study advances our understanding of synergistic agronomic management adaptation for sustaining soybean seed yields under warming, providing new quantitative evidence from the irrigated Soybeans in the US Mid-South that extends the existing literature. We found that soybean yields declined by 5.7% on average under rising temperatures. Adjusting the soybean maturity group, combined with early sowing, consistently prevented yield losses during high temperatures, resulting in an overall yield gain of 12% relative to the historical baseline.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Elvis F. Elli: Conceptualization, Data curation, Formal analysis, Funding acquisition, Visualization, Writing - original draft; Caio dos Santos: Formal analysis, Methodology, Software Writing - review & editing; Samuel B. Fernandes: Methodology, Software, Writing - review & editing; Montserrat Salmerón: Investigation, Methodology, Resources, Writing - review & editing; Rafael Battisti: Methodology, Writing - review & editing; Rogério de S. Nóia-Júnior: Conceptualization, Investigation, Writing - review & editing.

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SUPPLEMENTAL MATERIAL

Supplemental materials are included in this paper.

Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Figure S3. Environmental characterization of baseline conditions across all studied site-years. The APSIM phenology response curve (0-1) is presented, where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

Figure S4. USDA-NASS soybean sowing progress in Arkansas progress from 1990 to 2025. Grey lines represent the historical years, while the green, thicker line indicates the average sowing progress. 50% and 10% average sowing progress are indicated in the figure.

Figure S5. Density distribution of simulated soybean yields without adaptation under the baseline and +2°C scenarios with and without elevated [CO₂] (A), and its geographical distribution (B).

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For Review Only

Supplementary Material**SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION
SUSTAINS SOYBEAN YIELDS UNDER WARMING**

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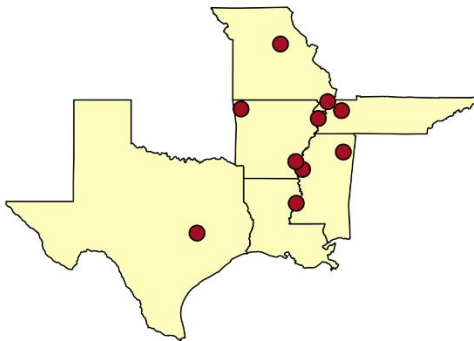
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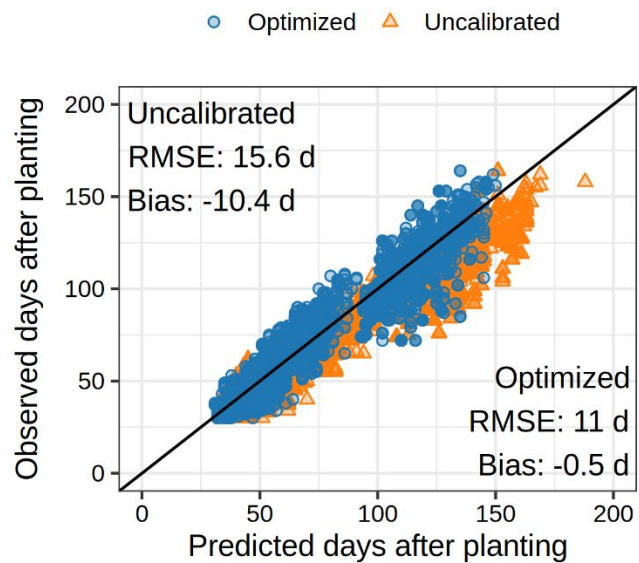
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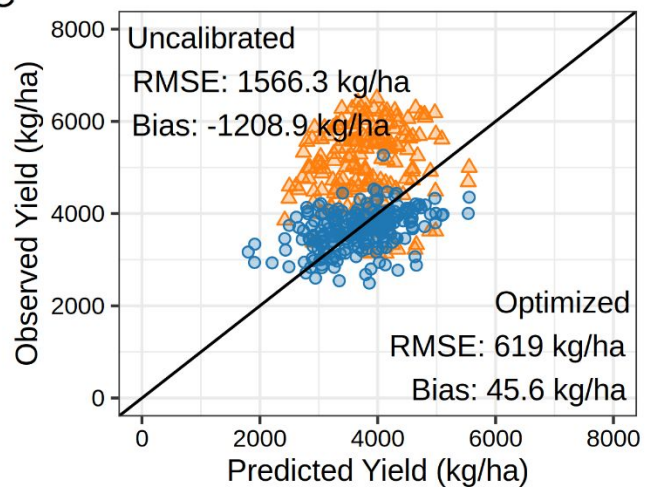


Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed days after sowing for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Model calibration

The occurrence of phenological stages, converted to days after sowing, was used to derive the cultivar-specific model parameters (Table S1) using an iterative optimization procedure based on the Nelder–Mead algorithm (Nelder and Mead, 1965). This algorithm explores the parameter space iteratively using a geometric figure called simplex. The optimization procedure aims to minimize the residual sums of squares (Wallach et al., 2001) between the estimated and actual timing to reach the studied phenology stages for each maturity group (early 4 and early 5). We ran the algorithm using R software (R Development Core Team, 2021) and the APSIMX package (Miguez, 2020). Our focus was on optimizing phenology parameters while keeping cardinal temperature-related parameters constant, giving little evidence of changes across genotypes (Archontoulis et al., 2014). Two growth parameters were selected: area of the largest leaf and radiation use efficiency. The optimization workflow followed an approach similar to that outlined in Wallach et al. (2019), in which all parameters were optimized simultaneously to potentially capture parameter interactions.

Table S1. Maturity group (MG) – specific default and optimized APSIM parameter values targeted in the optimization of early and late maturity groups 4 and 5 cultivars.

APSIM Parameter	Units	Early MG 4	Early MG 5
Vegetative Target	°C-day	484	634
Early Flowering Target	°C-day	185	189
Early Grain Filling Target	Fraction (0-1)	0.338	0.042
Late Grain Filling Target	°C-day	384	418
Reproductive Photoperiod Modifier	Fraction (0-1)	11.48, 17.24	12.70, 14.31
Area of Largest Leaf	m ²	0.004	0.004
Radiation Use Efficiency	g MJ ⁻¹	1.48	1.05

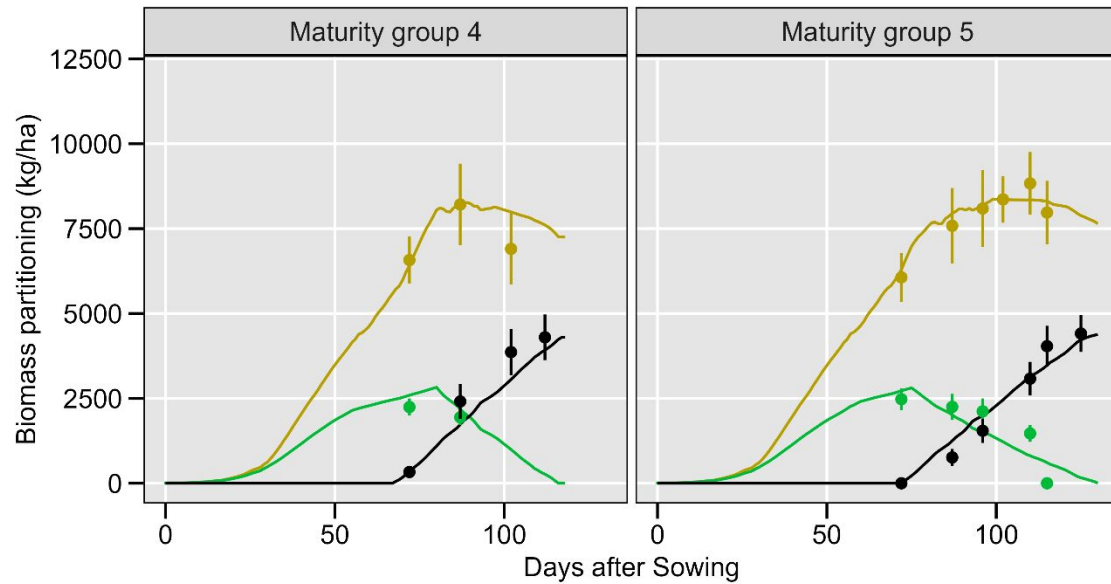


Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Study location and weather

The studied region is predominantly irrigated, with accumulated rainfall of 590 mm, unevenly distributed over the growing season (May to October). The average growing season temperature is 24°C, with values ranging from 20.9 to 27.1°C across all site-years (Figure S1). The studied area exhibits significant variability in soil conditions, ranging from shallow (less than 50 cm in depth) sandy soils with low soil water-holding capacity (0.4 mm cm⁻¹) to deeper soils with an effective root depth greater than 100 cm and high soil water-holding capacity (2 mm cm⁻¹). The soil organic carbon (0-30 cm) typically ranges from 0.5 to 3%.

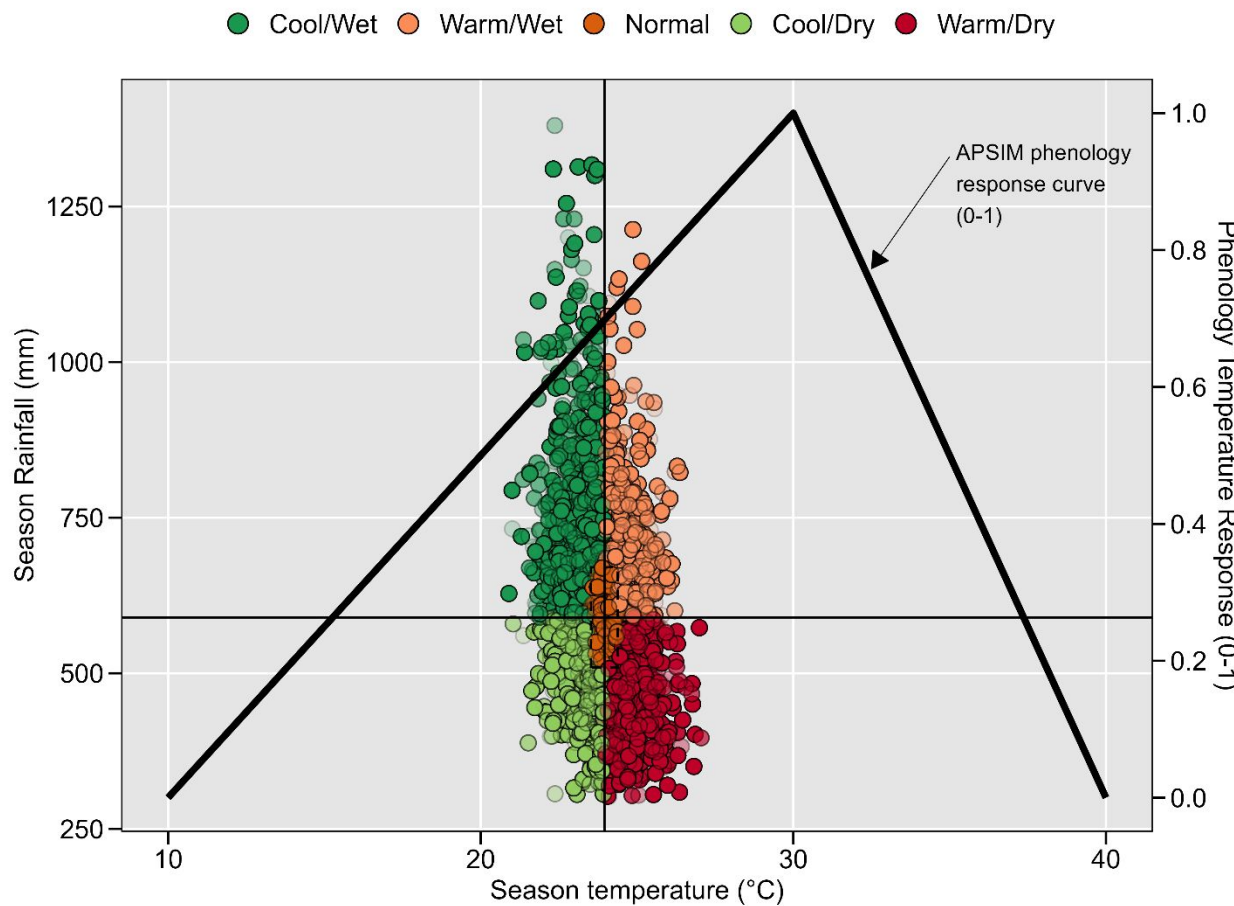


Figure S3. Environmental characterization of baseline conditions across all studied site-years. The APSIM phenology response curve (0-1) is presented, where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

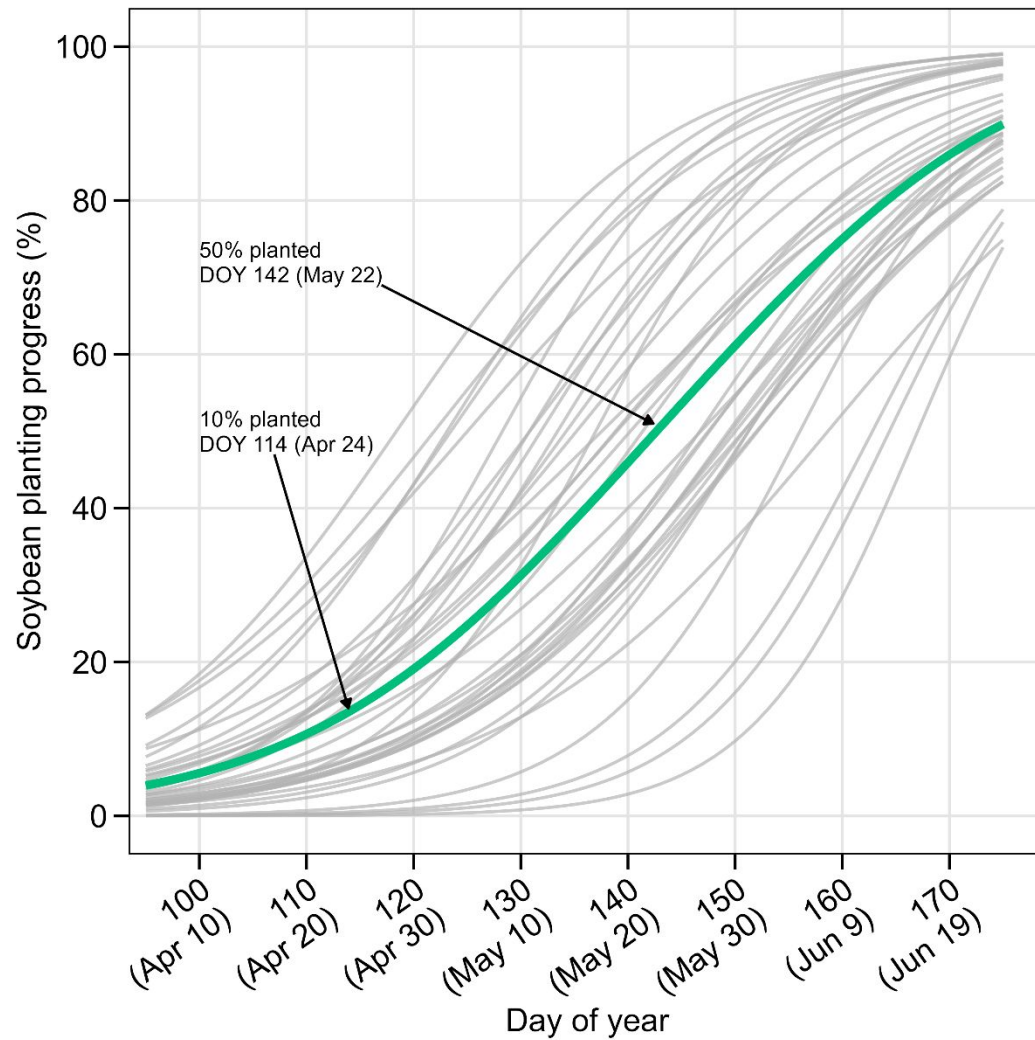


Figure S4. USDA-NASS soybean sowing progress in Arkansas progress from 1990 to 2025. Grey lines represent the historical years, while the green, thicker line indicates the average sowing progress. 50% and 10% average sowing progress are indicated in the figure.

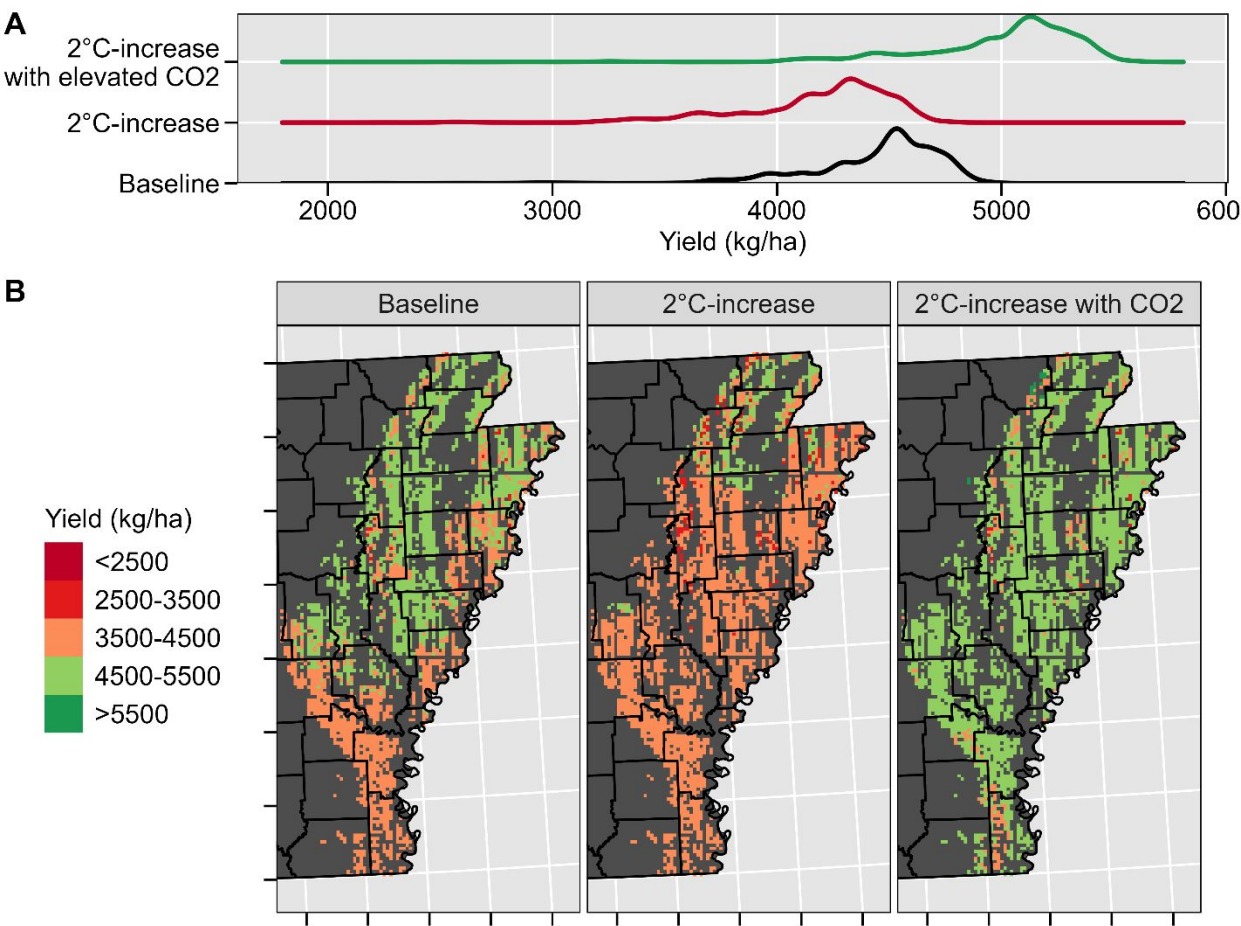


Figure S5. Density distribution of simulated soybean yields without adaptation under the baseline and +2°C scenarios with and without elevated [CO2] (A), and its geographical distribution (B).

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SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION
SUSTAINS SOYBEAN YIELDS UNDER WARMING
EARLIER PLANTING AND LATE-MATURING SOYBEANS OFFSET HIGH
TEMPERATURE STRESS

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ABSTRACT

High temperatures increasingly threaten U.S. soybean seed yields, yet actionable solutions remain limited. We evaluated the APSIM model against observed regional data from the U.S. Mid-South and conducted gridded simulations across 4,651 fields to identify how concurrent changes in planting date and maturity group affect yield under rising temperatures and elevated [CO₂]. The +2°C temperature scenario reduced yields by 5.7% on average. However, combining a later-maturing soybean genotype with earlier sowing synergistically increased soybean yield by 12% under the +2°C scenario. These results revealed an opportunity to sustain soybean yield through management and genotype selection under warming. Advancing our understanding of management and genotypic effects on soybean phenology and improving their representation in process-based models are therefore essential for developing adaptive strategies that can support durable yield gains under high-temperature conditions.

1 INTRODUCTION

Rising temperatures and changes in rainfall patterns can decrease U.S. soybean seed yields by 19% in the absence of adaptive strategies (Elli et al., 2022). Higher temperatures accelerate phenological development, shortening the crop cycle and limiting biomass accumulation and yield (Elli et al., 2022; Rose et al., 2016). Previous simulation studies indicate that adjusting soybean phenology may help offset yield reductions under global warming (Yang et al., 2025; Zabel et al., 2021). While projected increases in [CO₂] can partially offset heat-induced yield losses (Ainsworth et al., 2002a; Kothari et al., 2022), complementary agronomic adaptation may be needed in some environments to fully offset the yield penalty (Makowski et al., 2020). Yet, it remains unclear how concurrent alterations in soybean maturity group and sowing date will affect yields, and whether existing genotypic availability and planting date flexibility are sufficient to offset the negative effects of high temperatures. This study advances our understanding of how stacked agronomic management strategies influence soybean yield under elevated temperature in the U.S. Midsouth, a region particularly vulnerable to global warming. Maturity group selection and planting date are among the most actionable management decisions for farmers to adapt to rising temperatures (Thornton et al., 2014). A relatively simple and well-established strategy to adjust the soybean life cycle to the available growing season length is to select soybean maturity groups based on local temperature and photoperiod conditions (Mourtzinis and Conley, 2017). Moreover, earlier planting extends the growing season and shifts the crop cycle into cooler periods, reducing exposure to late-season drought and heat events (Edwards et al., 2003; Bastidas et al., 2008). While farmers have been shifting towards earlier planting and maturity groups well-adapted to a region, the synergistic effect of these shifts under a warming scenario remains unquantified. While this agronomic practice has

~~consistently increased soybean yield across the US, planting early is not always the best option, as there is an optimum window that maximizes capture of radiation resources and varies according to maturity group (Salmerón et al., 2016), indicating the need for cropping systems-specific recommendations.~~

We conducted a comprehensive simulation study across 4,651 fields and 40 weather-years using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. We considered soybean production systems in the Arkansas Delta region as a case study, an environment particularly vulnerable to warming (Sharma et al., 2023), with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025). We hypothesize that under high temperatures, using late-maturing soybean genotypes combined with early planting will offset the negative impacts of warming on soybean yield. Our objective was to quantify the individual and synergistic impacts of changes in maturity group and sowing date on adapting soybean production to high-temperature risks under current and elevated [CO₂].

2 MATERIAL AND METHODS

2.1. APSIM-Soybean model testing

The soybean model within APSIM Next Generation (Holzworth et al., 2018) was evaluated against regional data from the U.S. Midsouth covering multiple locations, planting dates, and cultivar maturity groups (Figure S1), as described in Salmerón et al. (2016). The APSIM-Soybean model has previously proven successful in simulating the growth of various soybean maturity groups under different environmental conditions in the US (Archontoulis et al., 2020; Elli et al., 2022). We further tested the model against experimental data and updated cultivar

coefficients to the latest APSIM Next Generation (Holzworth et al., 2018) version 2025.3.7681. Cultivar coefficients were calibrated for an early maturity group (MG) 4 (relative MG 4.0 to 4.4) and an early MG 5 (rMG from 5.0 to 5.4) cultivar. The calibration results confirmed that the model is robust across a wide range of environmental conditions (Figures S1 and S2 and Table S1), with root mean square errors within similar ranges to those reported in other studies for soybean phenology and yield (Archontoulis et al., 2014; Salmerón and Purcell, 2016; Salmerón et al., 2017).

2.2. Regional simulations and scenario analysis

We conducted a simulation exercise across 40 weather years and 4,651 fields in the US Mid-South. Further information on the weather conditions of the study area is presented in Figure S3.

We used the soybean cropland data layer (NASS, 2023) to identify soybean-cropped pixels and define the spatial extent of the simulation fields~~define the simulation fields~~, which were resampled to a 2.5 km grid ($\sim 0.025^\circ$)~~and clipped to the AR state boundary. The layer was reprojected to WGS84 to obtain latitude-longitude coordinates.~~ Each grid cell classified as soybean cropland was treated as one simulation field. We used the *apsimx* R package (Miguez, 2020) to run the simulations. Daily historical weather data on maximum and minimum temperatures, solar radiation, and rainfall were retrieved from the nearest weather station in the National Weather Service Network using the *apsimx* R package. [CO₂] was set at 350 ppm. Soil profile information was derived from SSURGO (Soil Survey Staff, 2023). Within each simulated grid cell, we used the most predominant soil profile. Soil water and inorganic nitrogen initial conditions were reset on January 1 each year and were held constant across locations to avoid confounding effects of soil vs. weather (Elli and Archontoulis, 2023). On the first day of the simulation (January 1), soil water content was set at field capacity, total inorganic nitrogen over

the soil profile was 50 kg N ha⁻¹, and maize residue on the soil surface was 2000 kg ha⁻¹ with a C:N ratio of 70.

For baseline simulations, the sowing date was May ~~15~~²², an average value for the studied region based on USDA-NASS 50% sowing progress from 1990 to 2025 (NASS, 2025; [Figure S4](#)). We used an early MG 4 cultivar, which is well adapted to the region (Salmerón et al., 2016; Salmerón et al., 2014). Soybean was irrigated as it is the predominant system across the studied region (NASS, 2025). Automatic irrigation was triggered between R1 and R7 when soil water reached 50% of the plant-available water in the 0-60 cm depth, to refill soil moisture to field capacity.

For the high-temperature scenario, we considered a 2°C increase (+2°C) in daily maximum and minimum temperatures, which is an average projection derived from 15 global circulation models for 2050, as reported by Elli et al. (2022). A +2°C scenario combined with an elevated [CO₂] to 540 ppm (Elli and Archontoulis, 2023) was also evaluated. We employed a simpler approach, rather than the direct outputs of global circulation models, because it has been shown to yield results with similar direction and magnitude while substantially reducing computational requirements (Elli et al., 2022). The adaptive scenarios included (1) an early MG 5 cultivar instead of early MG 4, (2) advancing sowing date by 12 days, corresponding to a realistic range within the upper tail of the NASS planting progress for the studied region (NASS, 2023), and (3) a combination of an early MG 5 cultivar with early sowing. Data analysis and visualization were conducted using R version 4.1.1 (R Development Core Team, 2024).

3 RESULTS

3.1. Elevated temperature and [CO₂] impacts on soybean yields without adaptation

The +2°C temperature scenario decreased soybean seed yields by 5.7% on average (~245 kg ha⁻¹), with values ranging from -14.7% to -2.5% across the 4,651 sites (Figure 1). The mean simulated yield for the baseline condition was 4,300 kg ha⁻¹ with a south-to-north geographical gradient ranging from 2,142 to 4,837 kg ha⁻¹ (Figure S5). Elevated [CO₂] consistently offset the negative effects of higher temperatures across sites (Figure 1), with an average yield 12.9% greater than under baseline conditions, and values ranging from 9.2% to 15.4% across all sites (Figure 1). A shortening of the total crop cycle was observed in 61% of the sites, with more pronounced changes in northern environments (Figure 2), where baseline temperatures are lower. The total crop cycle length decreased by up to 15 days across all site-years. The seed filling period was shortened by up to 9 days across all site-years.

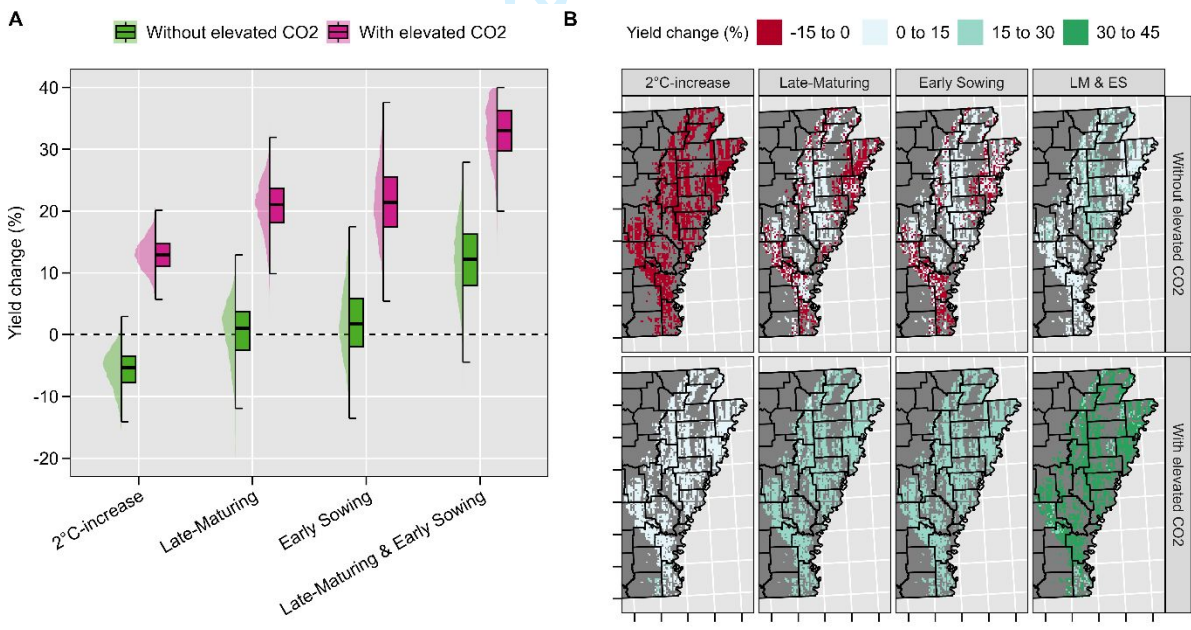
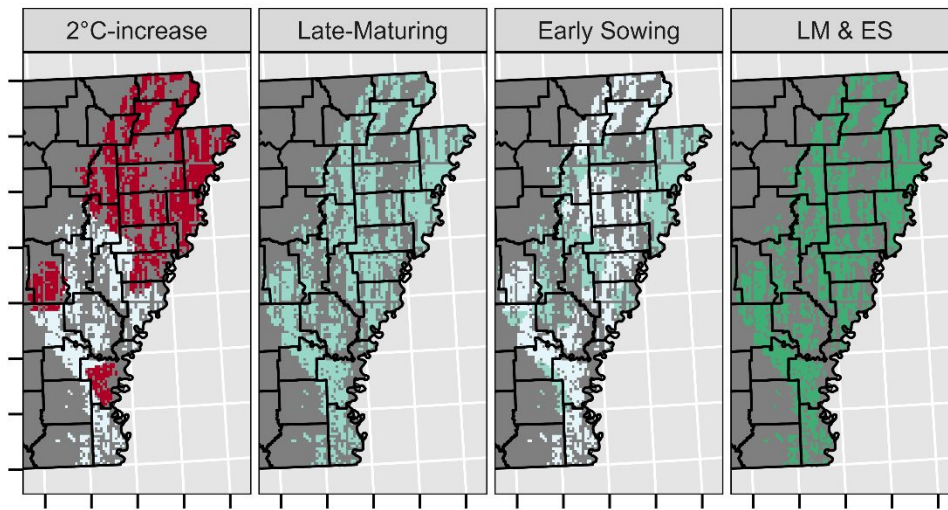


Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are

averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C. -

(A) Total crop cycle change (days) -8 to 0 0 to 8 8 to 15 15 to 22



(B) Seed-filling period change (days) -5 to 0 0 to 5 5 to 10 10 to 15

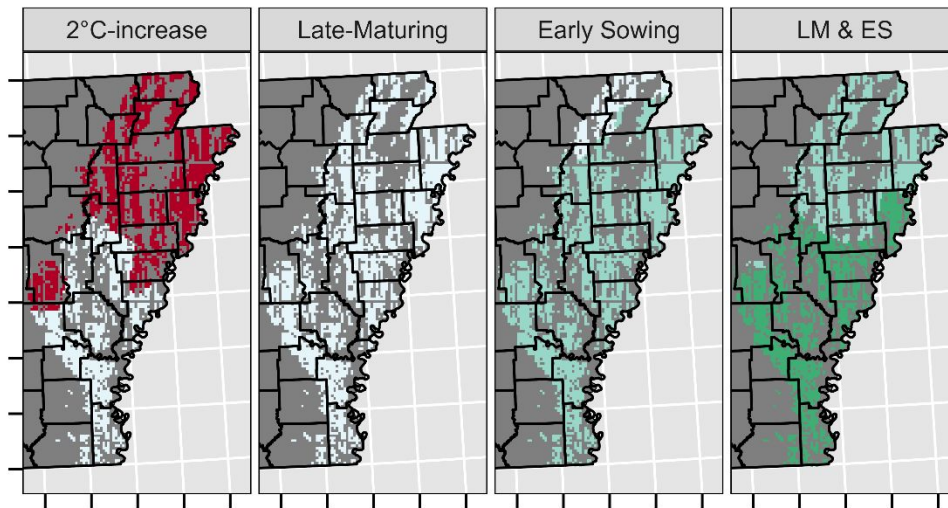


Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and

seed-filling period (B) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

3.2. Adaptive strategies to elevated temperature

Stacked agronomic management strategies were more effective than standalone strategies to maximize soybean yields under warming (Figure 1). Using a late-maturing soybean cultivar in combination with early sowing increased soybean seed yield by 12% under the +2°C temperature scenario, without elevated CO₂. The combined adaptive scenario extended the seed-filling period by an average of 10.4 days (Figure 2). Overall, the standalone strategy of using a late-maturing cultivar under a +2°C temperature scenario resulted in yields comparable to the historical baseline, with values ranging from -12.6% to 4.8% across fields. This practice increased yields at 62% of the 4,651 simulated sites, particularly in the central-north region (Figure 1B). The Early sowing strategy increased yields by an average of 2% relative to the baseline, with values ranging from -4.9% to 7.8%. This practice increased yields at 71% of the sites. Elevated [CO₂] altered the magnitude but not the direction of the yield response to stacked adaptive management options. The combined impact of elevated CO₂ and stacked agronomic management strategies increased yields by an average of 34%, with values ranging from 26.1% to 41.8% (Figure 1).

4 DISCUSSION

Our study revealed that soybean yields under warming can be sustained by concurrently adjusting cultivar maturity and planting date. Under the +2°C temperature scenario, the standalone strategies of using a late-maturing cultivar and planting earlier increased yield by an average of 0.5% and 2%, respectively, relative to the historical simulations. This implies that

individual strategies offset the impacts of higher temperature, resulting in yields comparable to baseline yields under current weather conditions. However, the offset was not consistent across all sites (Figure 1). The combined strategies increased seed yield by an average of 12% in relation to the baseline, with values ranging from 0.4% to 17.6% across sites, demonstrating a synergistic impact and emerging as the most promising approach to sustain soybean yield under a +2°C scenario. Our study extends previous work (Elli et al., 2022; Nendel et al., 2023) by quantifying the synergistic effect of stacked agronomic management strategies to mitigate heat risks in irrigated soybeans at a fine spatial resolution in the U.S. Mid-South.

We found a 5.7% decrease in soybean yield due to rising temperatures, with values ranging from -14.7% to -2.5% across sites (Figure 1). Elli et al. (2022) reported an average 19% reduction in seed yield at 10 locations in the US Corn Belt under a +2°C temperature scenario without adaptive strategies. While Elli et al. (2022) did not quantify the combined impact of high temperatures and stacked agronomic management, they found that sowing 12 days earlier increased soybean seed yield by 4.7%. Deryng et al. (2014) investigated the effects of heat stress on soybeans worldwide and predicted a 33% yield reduction by 2080. Nendel et al. (2023) projected that heat stress will increasingly outweigh drought stress as a constraint to soybean production in Europe.

While elevated [CO₂] can offset the negative impacts of rising temperatures (Ainsworth et al., 2002b), genotypic differences in [CO₂] response may exist (Bishop et al., 2015). Moreover, structural model uncertainties exist in predicting yield benefits under elevated [CO₂] (Kothari et al., 2022). Recognizing that soybeans are already a highly heat-tolerant crop, heat events above 28°C during the reproductive phase can reduce yield by lowering seed number and weight (Boote et al., 2005; Djanaguiraman et al., 2013). The APSIM-soybean version used in the

present study accounts for temperature effects on crop growth rate (Kothari et al., 2024), but does not account for the impacts of extreme heat on seed number determination, a common limitation in process-based models (Richetti et al., 2025). Moreover, APSIM assumes constant temperature responses throughout the life cycle, whereas in soybeans this response may change across developmental stages (Hatfield et al., 2011).

Warmer springs and longer frost-free periods have allowed farmers to plant earlier every year (Mourtzinis et al., 2019). However, this practice is not always feasible due to extreme rainfall events, which are projected to intensify further (Elli et al., 2022). Given the importance of adapting the soybean life cycle to temperature changes, further research is needed to enhance the fundamental understanding of soybean phenology and the genetic and management controls.

Extending the seed filling period was a promising strategy to maximize soybean seed yield (Figure 1), which agrees with the findings of Boote et al. (2003). However, phenology did not fully explain yield reduction under warming, especially in the Southern region (Figures 1 and 2). This could be associated with slower development on days when temperatures exceed the optimum temperature for soybean development in APSIM (30°C, Figure S3). Battisti et al. (2018) observed that, owing to the temperature response function governing phenological development, the APSIM model simulated a greater reduction in soybean cycle length under a +4.5°C warming scenario than under a +6.0°C scenario in Southern Brazil, with a mean air temperature 23.65°C. Further yield decreases were likely associated with high-temperature effects on radiation use efficiency (Kothari et al., 2022; Ruiz-Vera et al., 2013). While exploring optimum planting date windows for target environments was not the focus of this study, future research could investigate more extreme scenarios to enhance mechanistic understanding of yield responses under warming.

5 CONCLUSION

This study advances our understanding of synergistic agronomic management adaptation for sustaining soybean seed yields under warming, providing new quantitative evidence from the irrigated Soybeans in the US Mid-South that extends the existing literature. We found that soybean yields declined by 5.7% on average under rising temperatures. Adjusting the soybean maturity group, combined with early sowing, consistently prevented yield losses during high temperatures, resulting in an overall yield gain of 12% relative to the historical baseline.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Elvis F. Elli: Conceptualization, Data curation, Formal analysis, Funding acquisition, Visualization, Writing - original draft; Caio dos Santos: Formal analysis, Methodology, Software Writing - review & editing; Samuel B. Fernandes: Methodology, Software, Writing - review & editing; Montserrat Salmerón: Investigation, Methodology, Resources, Writing - review & editing; Rafael Battisti: Methodology, Writing - review & editing; Rogério de S. Nóia-Júnior: Conceptualization, Investigation, Writing - review & editing.

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develop climate-resilient cultivars). Sincere appreciation is extended to all researchers who collected the phenology data used for model testing.

SUPPLEMENTAL MATERIAL

Supplemental materials are included in this paper.

Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

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Figure S5. Density distribution of simulated soybean yields without adaptation under the baseline and +2°C scenarios with and without elevated [CO₂] (A), and its geographical distribution (B).

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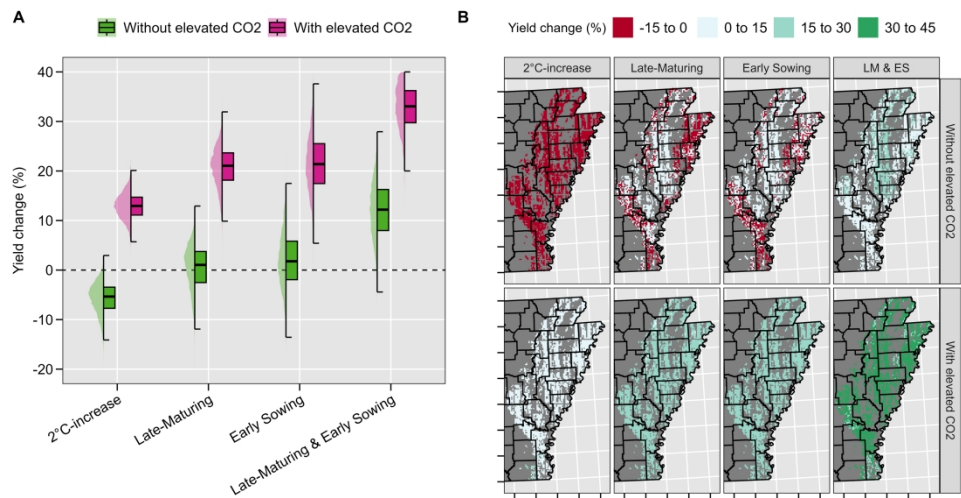
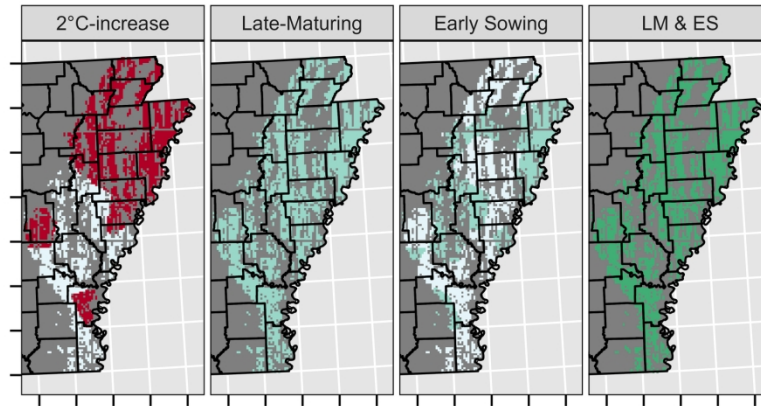


Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

299x149mm (300 x 300 DPI)

(A) Total crop cycle change (days) ■ -8 to 0 ■ 0 to 8 ■ 8 to 15 ■ 15 to 22



(B) Seed-filling period change (days) ■ -5 to 0 ■ 0 to 5 ■ 5 to 10 ■ 10 to 15

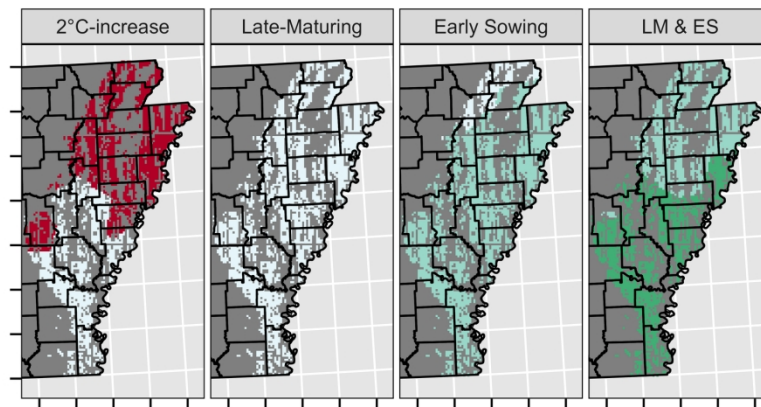


Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

179x179mm (300 x 300 DPI)